Vakint reduction benchmark

RustRed versus alphaLoop, MATAD and FMFT · reduction graphs from one to five loops · precision of the results

2026-11-13 · AMD EPYC 9754 128-Core Processor · one core per timed process except 11 marked cells · vakint 0.2.0

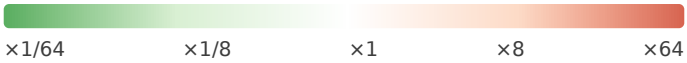
Reading guide. Bold: RustRed median, then the reference's (M = MATAD, F = FMFT, αL = alphaLoop) and the ratio RustRed ÷ reference; green = RustRed faster, red = slower (log scale); (·) = minimum over the runs. *Reduction*: masters retained; *full*: complete call with master values; *warm*: the identical call repeated in the process (step cache filled). Every timed call follows an untimed warm-up on another integral, so one-time setup is excluded (listed below the table). *IBP*: RustRed's time in the reduction itself. Five loops (no FORM backend): a master is recognised in any form and returned without a reduction walk; other inputs run RustRed's modular ordered walk (`modular_sampled`, on the cores a cell names).

Timings

Table 1: **RustRed versus the FORM backends** (MATAD and alphaLoop: one to three loops; FMFT: four loops; none reaches five). Terminals: distinct RustRed master symbols in the retained result (one to three loops: terminals of the closed program, symmetric copies included). Agreement: digits shared with RustRed at $m = \mu = 1$, worst order of ε.

Test case	Termi-nals	Reduction vs M / F	Reduction vs αL	Warm reduction	Full call vs M / F	Full call vs αL	Agree-ment
One loop							
QCD ladder, q = 0 quark loop, no rung	1	**5.52 ms**† (4.29) M 26.9 ms (26.3) **×0.20** IBP 1.91 ms	**5.52 ms**† (4.29) αL 27.8 ms (23.2) **×0.20**	**3.98 ms** (3.38) cold 5.52 ms† **×0.72**	**11.2 ms** (10.9) M 162 ms (157) **×1/14**	**11.2 ms** (10.9) αL 37.4 ms (34.8) **×0.30**	**31** digits
QCD ladder, O(q²) quark loop, no rung	1	**5.19 ms** (3.82) M 58.3 ms (45.7) **×1/11** IBP 1.60 ms	**5.19 ms** (3.82) αL 56.9 ms (44.5) **×1/11**	**5.41 ms** (5.19) cold 5.19 ms **×1.0**	**11.1 ms** (10.9) M 313 ms (306) **×1/28**	**11.1 ms** (10.9) αL 67.7 ms (62.3) **×0.16**	**31** digits
Tadpole T1 power 5 · rank 4	1	**4.79 ms** (3.35) M 24.7 ms (23.1) **×0.19** IBP 1.91 ms	**4.79 ms** (3.35) αL 29.2 ms (23.5) **×0.16**	**3.51 ms** (3.03) cold 4.79 ms **×0.73**	**10.1 ms** (7.92) M 157 ms (155) **×1/15**	**10.1 ms** (7.92) αL 34.5 ms (32.9) **×0.29**	**31** digits
Two loops							
QCD ladder, q = 0 1 gluon rung	2	**18.1 ms** (15.6) M 43.8 ms (40.4) **×0.41** IBP 3.36 ms	**18.1 ms** (15.6) αL 32.0 ms (30.9) **×0.56**	**15.9 ms** (15.8) cold 18.1 ms **×0.88**	**40.3 ms** (40.0) M 190 ms (188) **×0.21**	**40.3 ms** (40.0) αL 65.6 ms (57.6) **×0.61**	**31** digits
QCD ladder, O(q²) 1 gluon rung	2	**116 ms** (114) M 281 ms (277) **×0.41** IBP 7.61 ms	**116 ms** (114) αL 229 ms (222) **×0.51**	**122 ms** (120) cold 116 ms **×1.0**	**140 ms** (139) M 1.15 s (1.11) **×0.12**	**140 ms** (139) αL 293 ms (292) **×0.48**	**31** digits
Sunset T111 powers 3,2,2 · rank 4	2	**4.55 ms** (4.50) M 32.1 ms (31.1) **×0.14** IBP 1.09 ms	**4.55 ms** (4.50) αL 27.0 ms (26.8) **×0.17**	**4.29 ms** (4.20) cold 4.55 ms **×0.94**	**18.9 ms** (18.6) M 173 ms (172) **×0.11**	**18.9 ms** (18.6) αL 38.1 ms (35.5) **×0.50**	**31** digits
Three loops							
QCD ladder, q = 0 2 gluon rungs	14	**87.4 ms** (85.9) M 197 ms (189) **×0.44** IBP 60.2 ms	**87.4 ms** (85.9) αL 172 ms (169) **×0.51**	**47.4 ms** (47.1) cold 87.4 ms **×0.54**	**703 ms** (667) M 1.03 s (1.00) **×0.68**	**703 ms** (667) αL 277 ms (274) **×2.5**	**31** digits
QCD ladder, O(q²) 2 gluon rungs	14	**702 ms** (670) M 4.48 s (4.47) **×0.16** IBP 257 ms	**702 ms** (670) αL 5.03 s (5.00) **×0.14**	**522 ms** (513) cold 702 ms **×0.74**	**1.43 s** (1.23) M 12.1 s (11.8) **×0.12**	**1.43 s** (1.23) αL 5.38 s (5.17) **×0.27**	**31** digits
Tetrahedron D6 master	1	**7.84 ms** (7.55) M 77.2 ms (75.1) **×0.10** IBP 923 µs	**7.84 ms** (7.55) αL 30.2 ms (29.6) **×0.26**	**8.23 ms** (8.12) cold 7.84 ms **×1.0**	**18.8 ms** (18.3) M 269 ms (257) **×1/14**	**18.8 ms** (18.3) αL 54.7 ms (54.7) **×0.34**	**31** digits
Tetrahedron D6 3 dots · rank 4	38	**144 ms** (122) M 88.2 ms (85.7) **×1.6** IBP 134 ms	**144 ms** (122) αL 151 ms (143) **×0.95**	**71.5 ms** (66.1) cold 144 ms **×0.50**	**754 ms** (738) M 568 ms (553) **×1.3**	**754 ms** (738) αL 207 ms (205) **×3.6**	**31** digits
Four loops							
QCD ladder, q = 0 3 gluon rungs	10	**1.01 s** (1.00) F 6.51 s (6.24) **×0.16** IBP 275 ms	n/a	**822 ms** (784) cold 1.01 s **×0.81**	**2.51 s** (2.46) F 6.59 s (6.56) **×0.38**	n/a	**26** digits
QCD ladder, O(q²) 3 gluon rungs	10	**56.1 s**† (51.9) F 498 s (479) **×0.11** IBP 1.46 s	n/a	**51.5 s** (50.6) cold 56.1 s† **×0.92**	**57.1 s** (51.9) F 525 s (490) **×0.11**	n/a	**24** digits

Test case	Termi-nals	Reduction vs M / F	Reduction vs αL	Warm reduction	Full call vs M / F	Full call vs αL	Agree-ment
H family one line squared	15	**951 ms** (912) F 632 ms (620) **×1.5** IBP 871 ms	n/a	**431 ms** (428) cold 951 ms **×0.45**	**1.38 s** (1.30) F 658 ms (634) **×2.1**	n/a	**22** digits
FG family one line to the 4th	15	**241 ms** (233) F 427 ms (385) **×0.56** IBP 180 ms	n/a	**136 ms** (129) cold 241 ms **×0.56**	**2.90 s** (2.67) F 407 ms (395) **×7.1**	n/a	**23** digits
X family one line to the 4th	19	**4.80 s** (4.70) F 20.4 s (18.5) **×0.24** IBP 4.57 s	n/a	**3.00 s** (2.84) cold 4.80 s **×0.63**	**5.47 s** (5.45) F 18.6 s (18.5) **×0.29**	n/a	**21** digits
Five loops							
QCD ladder, q = 0 4 gluon rungs	79	**76.4 s**† 1 run no FORM backend on 8 cores	n/a	**100 s**† 1 run cold 76.4 s† **×1.3** on 8 cores	**117 s**† 1 run no FORM backend on 8 cores	n/a	—
QCD ladder, O(q²) 4 gluon rungs	79	**30.4 min**† 1 run no FORM backend on 8 cores	n/a	**16.3 min**† 1 run cold 30.4 min† **×0.54** on 8 cores	**22.8 min**† 1 run no FORM backend on 8 cores	n/a	—
Top-sector master (P105) 12 lines · scalar	1	**109 ms** (107) no FORM backend IBP 681 µs	n/a	**110 ms** (106) cold 109 ms **×1.0**	**125 ms** (109) no FORM backend	n/a	—
Dotted master (P63) 10 lines · rank 1	1	**110 ms** (105) no FORM backend IBP 652 µs	n/a	**103 ms** (102) cold 110 ms **×0.93**	**214 ms** (197) no FORM backend	n/a	—
Top sector (P105) master + one dot	98	**372 s**† 1 run no FORM backend on 8 cores	n/a	**288 s**† 1 run cold 372 s† **×0.77** on 8 cores	failed no FORM backend	n/a	—
Dotted-master sector (P63) master + one dot	79	**66.5 s**† 1 run no FORM backend on 8 cores	n/a	**29.1 s**† 1 run cold 66.5 s† **×0.44** on 8 cores	**72.7 s**† 1 run no FORM backend on 8 cores	n/a	—



† **Median not timed in an idle window** (the other cells' medians were, some after re-measurement): 1L QCD ladder, q = 0 · reduction: SMT sibling 100% busy; 4L QCD ladder, O(q²) · reduction: SMT sibling 39% busy; 5L QCD ladder, q = 0 · reduction: measured while the host was swapping (foreign + other jobs; 2026-11-12T20:33Z-20:42Z: MemAvailable min 92 GB, median swap-out 8,404 pages/s); not a clean timing, not comparable with earlier runs; SMT sibling 57% busy; other tasks used 17.8% of each of the 8 timed CPUs during the timed call; 5L QCD ladder, q = 0 · warm reduction: measured while the host was swapping (foreign + other jobs; 2026-11-12T20:33Z-20:42Z: MemAvailable min 92 GB, median swap-out 8,404 pages/s); not a clean timing, not comparable with earlier runs; SMT sibling 32% busy; other tasks used 31.0% of each of the 8 timed CPUs during the timed call; 5L QCD ladder, q = 0 · full call: measured while the host was swapping (foreign + other jobs; 2026-11-12T20:33Z-20:42Z: MemAvailable min 92 GB, median swap-out 8,404 pages/s); not a clean timing, not comparable with earlier runs; SMT sibling 46% busy; other tasks used 22.0% of each of the 8 timed CPUs during the timed call; 5L QCD ladder, O(q²) · reduction: measured while the host was swapping (foreign + other jobs; 2026-11-12T21:11Z-22:41Z: MemAvailable min 78 GB, median swap-out 30,606 pages/s); not a clean timing, not comparable with earlier runs; SMT sibling 96% busy; other tasks used 13.4% of each of the 8 timed CPUs during the timed call; 5L QCD ladder, O(q²) · warm reduction: measured while the host was swapping (foreign + other jobs; 2026-11-12T21:11Z-22:41Z: MemAvailable min 78 GB, median swap-out 30,606 pages/s); not a clean timing, not comparable with earlier runs; SMT sibling 83% busy; other tasks used 17.9% of each of the 8 timed CPUs during the timed call; 5L QCD ladder, O(q²) · full call: measured while the host was swapping (foreign + other jobs; 2026-11-12T21:11Z-22:41Z: MemAvailable min 78 GB, median swap-out 30,606 pages/s); not a clean timing, not comparable with earlier runs; SMT sibling 89% busy; other tasks used 26.1% of each of the 8 timed CPUs during the timed call; 5L Top sector · reduction: measured while the host was swapping (foreign + other jobs; 2026-11-12T20:49Z-21:18Z: MemAvailable min 105 GB, median swap-out 10,207 pages/s); not a clean timing, not comparable with earlier runs; SMT sibling 28% busy; other tasks used 9.5% of each of the 8 timed CPUs during the timed call; 5L Top sector · warm reduction: measured while the host was swapping (foreign + other jobs; 2026-11-12T20:49Z-21:18Z: MemAvailable min 105 GB, median swap-out 10,207 pages/s); not a clean timing, not comparable with earlier runs; SMT sibling 25% busy; other tasks used 11.2% of each of the 8 timed CPUs during the timed call; 5L Dotted-master sector · reduction: measured while the host was swapping (foreign + other jobs; 2026-11-12T21:20Z-21:26Z: MemAvailable min 94 GB, median swap-out 31,960 pages/s); not a clean timing, not comparable with earlier runs; other tasks used 12.1% of each of the 8 timed CPUs during the timed call; 5L Dotted-master sector · warm reduction: measured while the host was swapping (foreign + other jobs; 2026-11-12T21:20Z-21:26Z: MemAvailable min 94 GB, median swap-out 31,960 pages/s); not a clean timing, not comparable with earlier runs; other tasks used 11.1% of each of the 8 timed CPUs during the timed call; 5L Dotted-master sector · full call: measured while the host was swapping (foreign + other jobs; 2026-11-12T21:20Z-21:26Z: MemAvailable min 94 GB, median swap-out 31,960 pages/s); not a clean timing, not comparable with earlier runs; other tasks used 6.1% of each of the 8 timed CPUs during the timed call.

Once per process (excluded from every cell; first-call setup = first minus second call on the warm-up integral)

	1L	2L	3L	4L	5L
RustRed data load	108 ms	96.1 ms	94.1 ms	5.96 s	37.8 s
RustRed full call	2.60 ms	5.29 ms	864 ms	87.5 ms	614 ms
MATAD full call	85.3 ms	82.9 ms	82.0 ms	—	—
alphaLoop full call	44.0 ms	43.9 ms	45.1 ms	—	—
FMFT full call	—	—	—	64.6 ms	—

Failed calls.
5L Top sector (P105), master + one dot · full call (1/1 runs): vakint fails closed (MissingMasterOrders): 72 masters lack Laurent orders this result needs (the reduction coefficients have poles at d = 4; deepest orders needed: 5L to ε^1, 4L to ε^4, 3L to ε^3, 1L to ε^6), so the result fails from ε^−2 on

Provenance

Data (timings)	vakint-external-data `v0.2.3` · commit `1be01449faa1` · root manifest `d2a1c58217e732f7`
Vakint	0.2.0 · release build of `da7e31d5794a` (clean) · harness `980f0d0e3247d441` · FORM 5.0.0
Host	AMD EPYC 9754 128-Core Processor, 384 logical CPUs, shared; timed on 25 pinned CPUs in 5 L3 domains (listed in suite.json)
Contention	314 samples, 30 still contended after re-measurement († thresholds per timed CPU: SMT sibling 25% busy, other tasks 3%, call run-queue wait 3% when other tasks ran) · medians (max): SMT sibling busy 4.1% (100%), timed call in the run queue 0% (9%), other tasks per timed CPU 0.5% (27.2%) · 1-min load ≤ 332.7
Regenerate	`crates/vakint/docs/benchmark/make-benchmark.sh DATA_ROOT BINARY OUT --install` after a data release `DATA_ROOT - OUT --refresh-tables --install` (timings kept) · assembled 2026-11-13T03:03:03Z

Reduction graphs

Nodes are the vacuum topologies of the shipped reduction, on the row of their number of lines; an edge removes one line (thicker: several lines), as in Fig. 21 of B. Ruijl's thesis (Leiden, 2017). Products (one-vertex-reducible graphs) are merged by their factors; numbers count masters (one to three loops: RustRed terminals).

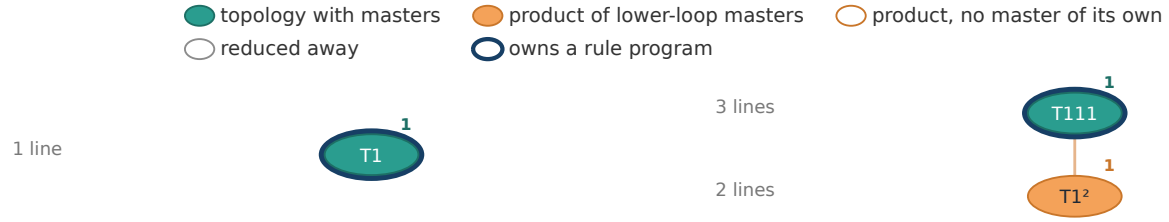


Figure 1: **1-loop reduction graph.** T1 carries the one-loop master; dots and numerators reduce inside it. 1 topology · 0 edges · 1 terminal.

Figure 2: **2-loop reduction graph.** Removing any sunset line gives T1². 2 topologies · 1 edge · 2 terminals.

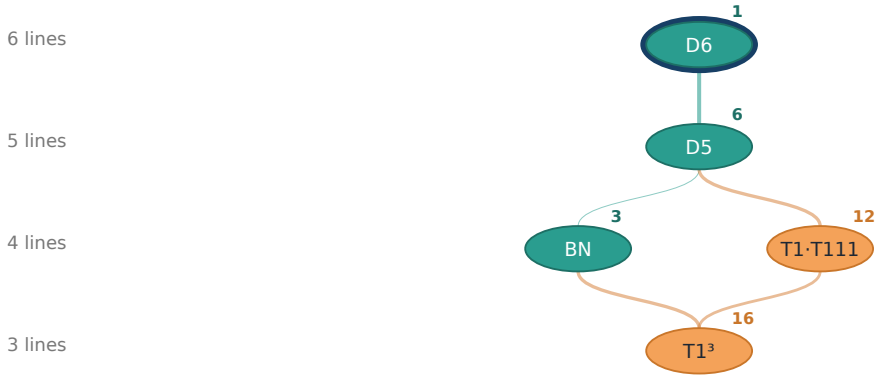


Figure 3: **3-loop reduction graph.** MATAD names; symmetric sectors are separate terminals of the closed program. 5 topologies · 5 edges · 38 terminals.

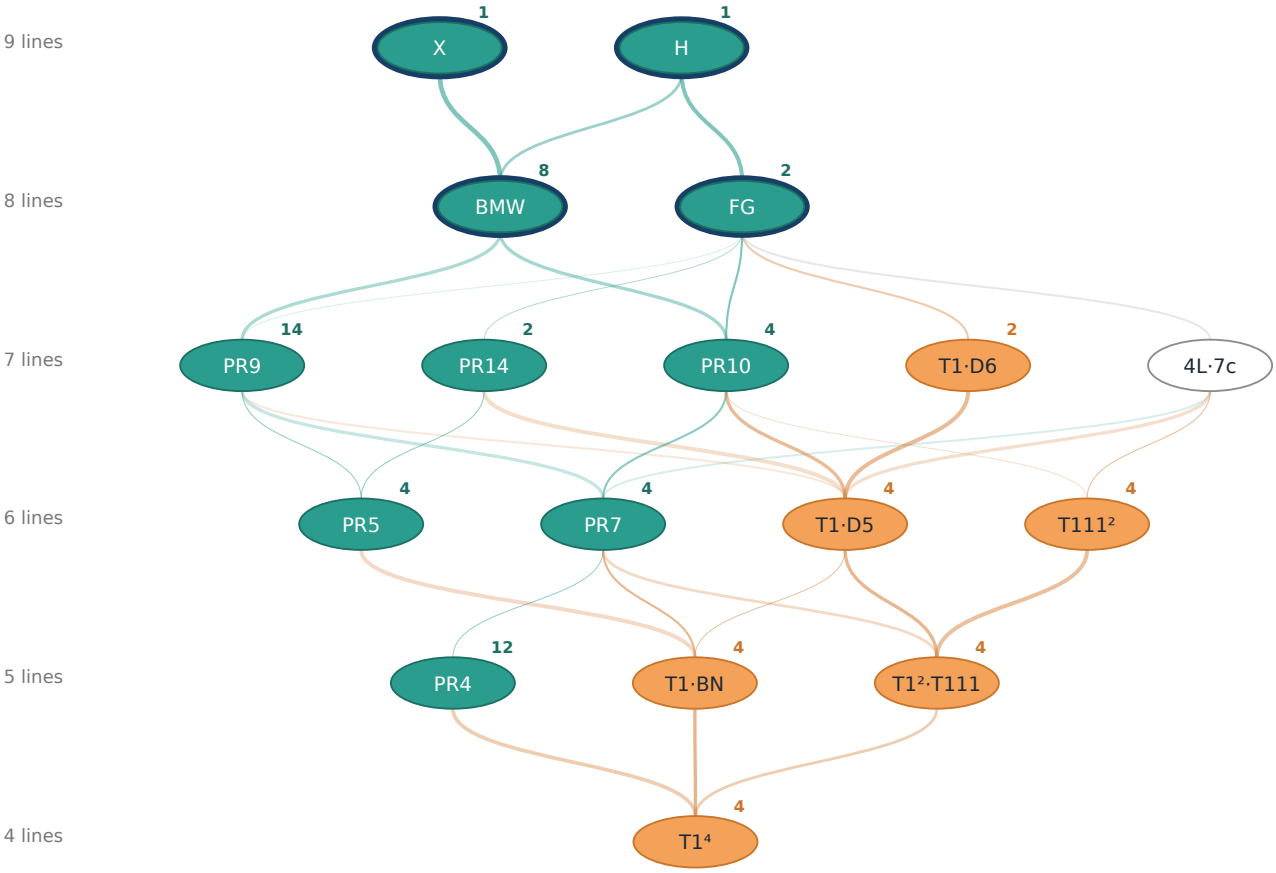


Figure 4: **4-loop reduction graph.** Rings: the FMFT family programs H, X, BMW, FG; irreducible topologies named by their FMFT master PRk. 17 topologies · 32 edges · 74 masters.

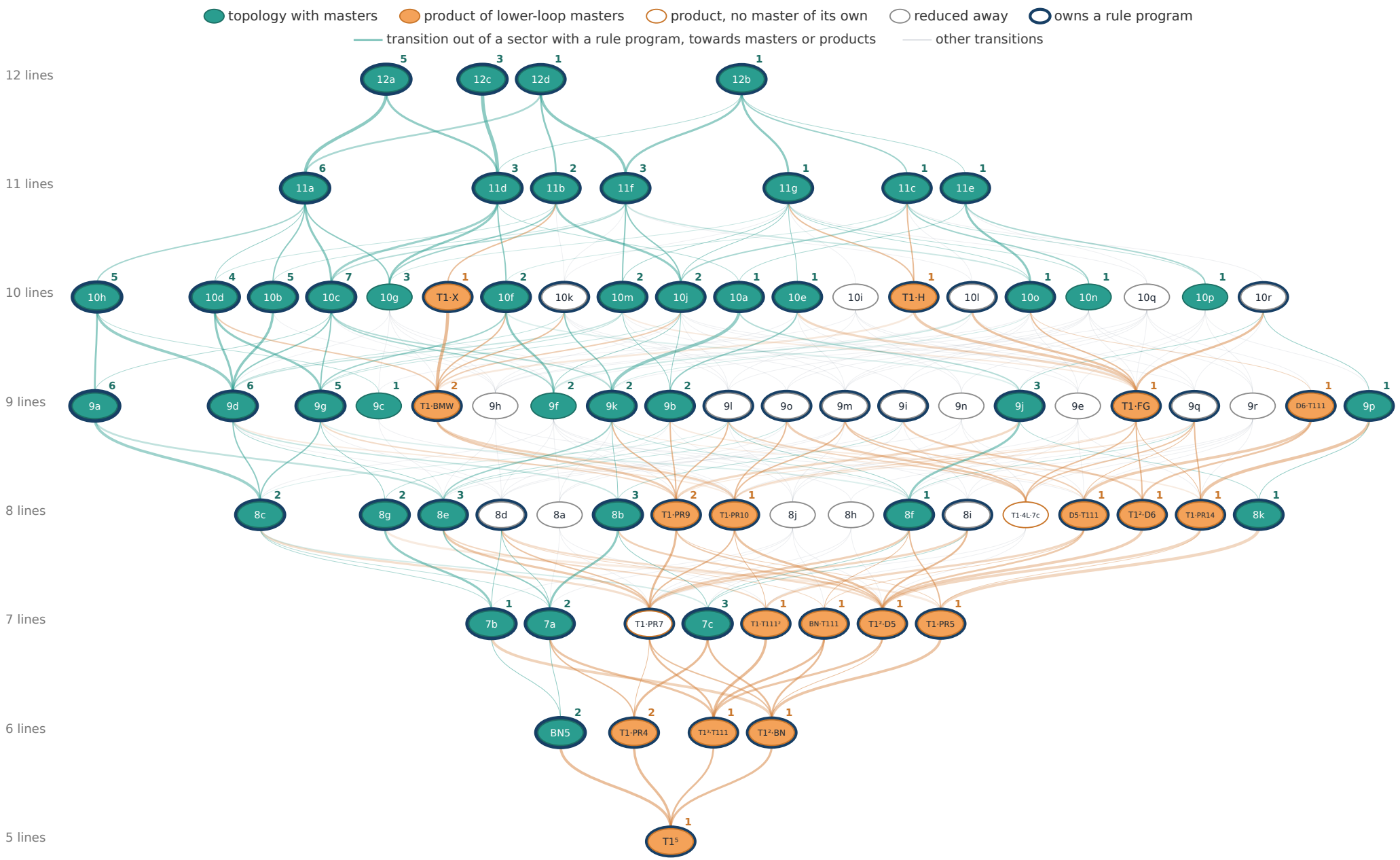


Figure 5: **5-loop reduction graph.** Rings: the 67 owner sectors of the saved application (width grows with their rule tables); names: lines and a letter. Benchmark masters: P105 in 12b, P63 in 10c. 82 topologies · 311 edges · 131 masters.

Precision of the masters and of the results

FINAL · data tree `/shared/vhirschi/t3code/vakint_dev/p_integral_dev/wt/bench-refresh-xd` (git `1be01449faa1`, tree `e1fefefae5d0`) · v0.2.3 root manifest `d2a1c58217e732f7` · 4L constants `1338cfe501ffb516` · 5L values build 2026-11-10T07:24:14Z (producer-update `7123159fa77a963d`) · generated 2026-11-13T03:03:07Z · tables `80266da44069c734`

Class of every Laurent coefficient, worst case X rounded down. **Precision table 1**: every master the data tree ships, at every order it holds. **Precision table 2**: the result of any single-scale vacuum input with L loops (RustRed backend), worst case over the inputs; an order that needs master coefficients the data do not hold (spurious poles of the reduction, lower-loop factors of 5L products) is not available.

S exact, periods at any precision · **S*(X)** exact, periods to at most X digits · **N(X)** numerical only, at most X digits · — not available

Precision table 1. Master coefficients — every master of the data tree (rows, by loop count; 5L: the 640-pack name in brackets) at every Laurent order held (columns); — = not held. [p] X rests in part on a constant computed with two profiles of one code beyond its independent confirmation; [n] Numerical mode returns more digits.

master	ε^{-5}	ε^{-4}	ε^{-3}	ε^{-2}	ε^{-1}	ε^{0}	ε^{1}	ε^{2}	ε^{3}	ε^{4}	ε^{5}
1L · 1 master · 7 S, 0 S*, 0 N											
1L/P1	—	—	—	—	S	S	S	S	S	S	S
2L · 2 masters · 13 S, 0 S*, 0 N											
2L/P1	—	—	—	S	S	S	S	S	S	—	—
2L/P2	—	—	—	S	S	S	S	S	S	S	—
3L · 38 masters · 252 S, 0 S*, 0 N											
3L/P1	—	—	S	S	S	S	S	S	—	—	—
3L/P2	—	—	S	S	S	S	S	S	—	—	—
3L/P3	—	—	S	S	S	S	S	S	—	—	—
3L/P4	—	—	S	S	S	S	S	S	S	—	—
3L/P5	—	—	S	S	S	S	S	S	—	—	—
3L/P6	—	—	S	S	S	S	S	S	—	—	—
3L/P7	—	—	S	S	S	S	S	S	—	—	—
3L/P8	—	—	S	S	S	S	S	S	S	—	—
3L/P9	—	—	S	S	S	S	S	S	—	—	—
3L/P10	—	—	S	S	S	S	S	S	S	—	—
3L/P11	—	—	S	S	S	S	S	S	—	—	—
3L/P12	—	—	S	S	S	S	S	S	S	—	—
3L/P13	—	—	S	S	S	S	S	S	S	S	—
3L/P14	—	—	S	S	S	S	S	S	S	—	—
3L/P15	—	—	S	S	S	S	S	S	—	—	—
3L/P16	—	—	S	S	S	S	S	S	—	—	—
3L/P17	—	—	S	S	S	S	S	S	—	—	—
3L/P18	—	—	S	S	S	S	S	S	S	—	—
3L/P19	—	—	S	S	S	S	S	S	—	—	—
3L/P20	—	—	S	S	S	S	S	S	—	—	—
3L/P21	—	—	S	S	S	S	S	S	S	S	—
3L/P22	—	—	S	S	S	S	S	S	S	—	—
3L/P23	—	—	S	S	S	S	S	S	S	—	—
3L/P24	—	—	S	S	S	S	S	S	S	—	—
3L/P25	—	—	S	S	S	S	S	S	—	—	—
3L/P26	—	—	S	S	S	S	S	S	S	—	—
3L/P27	—	—	S	S	S	S	S	S	—	—	—
3L/P28	—	—	S	S	S	S	S	S	S	S	—
3L/P29	—	—	S	S	S	S	S	S	S	—	—
3L/P30	—	—	S	S	S	S	S	S	S	—	—
3L/P31	—	—	S	S	S	S	S	S	—	—	—
3L/P32	—	—	S	S	S	S	S	S	S	—	—
3L/P33	—	—	S	S	S	S	S	S	S	—	—
3L/P34	—	—	S	S	S	S	S	S	S	—	—
3L/P35	—	—	S	S	S	S	S	S	S	—	—
3L/P36	—	—	S	S	S	S	S	S	S	—	—
3L/P37	—	—	S	S	S	S	S	S	S	—	—
3L/P38	—	—	S	S	S	S	S	S	—	—	—
4L · 74 masters · 376 S, 92 S*, 0 N											
4L/P1	—	S	S	S	S	S	S	S	—	—	—
4L/P2	—	S	S	S	S	S	S	S	S	—	—
4L/P3	—	S	S	S	S	S	S	S	—	—	—
4L/P4	—	S	S	S	S	S	S	S	—	—	—
4L/P5	—	S	S	S	S	S	S	—	—	—	—
4L/P6	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P7	—	S	S	S	S	S*(20000)	S*(5542)	—	—	—	—
4L/P8	—	S	S	S	S	S*(5527)	S*(5525)	—	—	—	—
4L/P9	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—

master	ε^{-5}	ε^{-4}	ε^{-3}	ε^{-2}	ε^{-1}	ε^{0}	ε^{1}	ε^{2}	ε^{3}	ε^{4}	ε^{5}
4L/P10	—	S	S	S	S	S*(5527)	S*(5525)	—	—	—	—
4L/P11	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P12	—	S	S	S	S	S	S	S	—	—	—
4L/P13	—	S	S	S	S	S*(5542)	S*(5406)	—	—	—	—
4L/P14	—	S	S	S	S	S*(5358)	S*(5357)[p]	—	—	—	—
4L/P15	—	S	S	S	S	S*(5542)	S*(5527)	—	—	—	—
4L/P16	—	S	S	S	S	S*(5542)	S*(5527)	—	—	—	—
4L/P17	—	S	S	S	S	S*(5363)[n]	S*(5358)	—	—	—	—
4L/P18	—	S	S	S	S	S*(2395)[p]	S*(2395)[p]	—	—	—	—
4L/P19	—	S	S	S	S	S*(5363)[n]	S*(5361)	—	—	—	—
4L/P20	—	S	S	S	S	S	S	S	—	—	—
4L/P21	—	S	S	S	S	S	S	S	S	—	—
4L/P22	—	S	S	S	S	S	S	S	—	—	—
4L/P23	—	S	S	S	S	S	S	S	—	—	—
4L/P24	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P25	—	S	S	S	S	S*(20000)	S*(5542)	—	—	—	—
4L/P26	—	S	S	S	S	S	S	S	—	—	—
4L/P27	—	S	S	S	S	S*(5542)	S*(5406)	—	—	—	—
4L/P28	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P29	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P30	—	S	S	S	S	S	S	—	—	—	—
4L/P31	—	S	S	S	S	S	S	—	—	—	—
4L/P32	—	S	S	S	S	S*(5527)	S*(5525)	—	—	—	—
4L/P33	—	S	S	S	S	S	S	—	—	—	—
4L/P34	—	S	S	S	S	S*(2805)[p]	S*(2601)[p]	—	—	—	—
4L/P35	—	S	S	S	S	S*(5542)	S*(5527)	—	—	—	—
4L/P36	—	S	S	S	S	S	S	S	—	—	—
4L/P37	—	S	S	S	S	S	S	S	S	—	—
4L/P38	—	S	S	S	S	S	S	S	—	—	—
4L/P39	—	S	S	S	S	S	S	S	—	—	—
4L/P40	—	S	S	S	S	S	S	—	—	—	—
4L/P41	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P42	—	S	S	S	S	S*(20000)	S*(5542)	—	—	—	—
4L/P43	—	S	S	S	S	S*(5527)	S*(5525)	—	—	—	—
4L/P44	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P45	—	S	S	S	S	S*(5527)	S*(5525)	—	—	—	—
4L/P46	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P47	—	S	S	S	S	S	S	S	—	—	—
4L/P48	—	S	S	S	S	S*(5542)	S*(5406)	—	—	—	—
4L/P49	—	S	S	S	S	S*(5358)	S*(5357)[p]	—	—	—	—
4L/P50	—	S	S	S	S	S*(5542)	S*(5527)	—	—	—	—
4L/P51	—	S	S	S	S	S*(5542)	S*(5527)	—	—	—	—
4L/P52	—	S	S	S	S	S*(5363)[n]	S*(5358)	—	—	—	—
4L/P53	—	S	S	S	S	S	S	S	—	—	—
4L/P54	—	S	S	S	S	S	S	S	S	—	—
4L/P55	—	S	S	S	S	S	S	S	—	—	—
4L/P56	—	S	S	S	S	S	S	S	—	—	—
4L/P57	—	S	S	S	S	S	S	—	—	—	—
4L/P58	—	S	S	S	S	S	S	—	—	—	—
4L/P59	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P60	—	S	S	S	S	S*(20000)	S*(5542)	—	—	—	—
4L/P61	—	S	S	S	S	S	S	S	—	—	—
4L/P62	—	S	S	S	S	S*(5542)	S*(5406)	—	—	—	—
4L/P63	—	S	S	S	S	S*(5527)	S*(5525)	—	—	—	—

master	ε^{-5}	ε^{-4}	ε^{-3}	ε^{-2}	ε^{-1}	ε^{0}	ε^{1}	ε^{2}	ε^{3}	ε^{4}	ε^{5}
4L/P64	—	S	S	S	S	S	S	—	—	—	—
4L/P65	—	S	S	S	S	S*(2805)[p]	S*(2601)[p]	—	—	—	—
4L/P66	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P67	—	S	S	S	S	S*(5542)	S*(5527)	—	—	—	—
4L/P68	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P69	—	S	S	S	S	S*(5358)	S*(5357)[p]	—	—	—	—
4L/P70	—	S	S	S	S	S*(2600)[p]	S*(2348)[p]	—	—	—	—
4L/P71	—	S	S	S	S	S*(5542)	S*(5527)	—	—	—	—
4L/P72	—	S	S	S	S	S*(5527)	S*(5525)	—	—	—	—
4L/P73	—	S	S	S	S	S*(5363)[n]	S*(5361)	—	—	—	—
4L/P74	—	S	S	S	S	S*(5363)[n]	S*(5358)	—	—	—	—
5L · 131 masters · 562 S, 29 S*, 195 N											
5L/P1 (P21)	S	S	S	N(67)	N(58)	N(59)	—	—	—	—	—
5L/P2 (P29)	S	S	S	N(83)	N(83)	N(75)	—	—	—	—	—
5L/P3 (P33)	S	S	S	N(69)	N(64)	N(61)	—	—	—	—	—
5L/P4 (P37)	S	S	S	N(65)	N(58)	N(56)	—	—	—	—	—
5L/P5 (P49)	S	S	S	N(65)	N(58)	N(57)	—	—	—	—	—
5L/P6 (P60)	S	S	N(65)	N(65)	N(57)	N(57)	—	—	—	—	—
5L/P7 (P65)	S	S	S	S	S	S	—	—	—	—	—
5L/P8 (P67)	S	S	S	S	S	N(142)	—	—	—	—	—
5L/P9 (P69)	S	S	S	S	S	N(142)	—	—	—	—	—
5L/P10 (P73)	S	S	S	N(59)	N(57)	N(56)	—	—	—	—	—
5L/P11 (P76)	S	S	S	S	S	S	—	—	—	—	—
5L/P12	S	S	S	S	S	S	—	—	—	—	—
5L/P13	S	S	S	S	S	S	—	—	—	—	—
5L/P14 (P95)	S	S	S	S	S	N(57)	—	—	—	—	—
5L/P15 (P100)	S	S	S	S	S	N(59)	—	—	—	—	—
5L/P16 (P101)	S	S	S	S	S	S	—	—	—	—	—
5L/P17 (P111)	S	S	S	N(59)	N(56)	N(56)	—	—	—	—	—
5L/P18 (P115)	S	S	S	N(65)	N(62)	N(58)	—	—	—	—	—
5L/P19 (P124)	S	S	S	S	N(57)	N(58)	—	—	—	—	—
5L/P20 (P126)	S	S	S	S	N(65)	N(61)	—	—	—	—	—
5L/P21 (P129)	S	S	S	S	S*(5542)	S*(5406)	—	—	—	—	—
5L/P22 (P131)	S	S	S	N(65)	N(57)	N(56)	—	—	—	—	—
5L/P23 (P134)	S	S	S	S	N(84)	N(75)	—	—	—	—	—
5L/P24 (P140)	S	S	S	S	N(57)	N(57)	—	—	—	—	—
5L/P25 (P142)	S	S	S	S	S	N(60)	—	—	—	—	—
5L/P26 (P144)	S	S	S	S	S*(5363)	N(57)	—	—	—	—	—
5L/P27 (P158)	S	S	S	S	S	S	—	—	—	—	—
5L/P28 (P159)	S	S	S	S	S	S	—	—	—	—	—
5L/P29 (P169)	S	S	S	S	N(56)	N(56)	—	—	—	—	—
5L/P30 (P170)	S	S	S	S	S	N(47)	—	—	—	—	—
5L/P31 (P171)	S	S	S	S	S	N(48)	—	—	—	—	—
5L/P32 (P172)	S	S	S	S	S	N(47)	—	—	—	—	—
5L/P33 (P178)	S	S	S	S	N(83)	N(75)	—	—	—	—	—
5L/P34 (P179)	S	S	S	S	N(83)	N(75)	—	—	—	—	—
5L/P35 (P182)	S	S	S	S	S*(2805)[p]	S*(2601)[p]	—	—	—	—	—
5L/P36 (P202)	S	S	S	S	S	N(69)	—	—	—	—	—
5L/P37 (P204)	S	S	S	S	S	N(56)	—	—	—	—	—
5L/P38 (P238)	S	S	S	S	N(56)	N(55)	—	—	—	—	—
5L/P39 (P245)	S	S	S	S	S*(5363)	N(57)	—	—	—	—	—
5L/P40 (P263)	S	S	S	S	N(55)	N(56)	—	—	—	—	—
5L/P41 (P264)	S	S	S	S	S	N(43)	—	—	—	—	—
5L/P42 (P299)	S	S	S	S	S	N(58)	—	—	—	—	—
5L/P43 (P312)	S	S	S	S	N(57)	N(57)	—	—	—	—	—
5L/P44 (P314)	S	S	S	S	S	N(74)	—	—	—	—	—
5L/P45 (P315)	S	S	S	S	S	N(66)	—	—	—	—	—
5L/P46 (P321)	S	S	S	S	N(57)	N(58)	—	—	—	—	—
5L/P47 (P352)	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—	—
5L/P48 (P353)	S	S	S	S	S	S	—	—	—	—	—
5L/P49 (P356)	S	S	S	S	S	S	—	—	—	—	—
5L/P50 (P357)	S	S	S	N(47)	N(47)	N(47)	—	—	—	—	—
5L/P51 (P359)	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—	—
5L/P52 (P361)	S	S	S	N(65)	N(57)	N(57)	—	—	—	—	—
5L/P53 (P367)	S	S	S	S	N(83)	N(75)	—	—	—	—	—
5L/P54 (P368)	S	S	S	S	N(82)	N(76)	—	—	—	—	—
5L/P55 (P369)	S	S	S	S	S*(5527)	S*(5525)	—	—	—	—	—
5L/P56 (P370)	S	S	S	S	N(82)	N(74)	—	—	—	—	—
5L/P57 (P375)	S	S	S	S	N(82)	N(76)	—	—	—	—	—
5L/P58 (P379)	S	S	S	S	S	N(74)	—	—	—	—	—
5L/P59 (P382)	S	S	S	S	S*(5358)	S*(5357)[p]	—	—	—	—	—
5L/P60 (P383)	S	S	S	N(65)	N(57)	N(57)	—	—	—	—	—
5L/P61 (P384)	S	S	S	S	S	N(57)	—	—	—	—	—
5L/P62 (P385)	S	S	S	S	S	N(56)	—	—	—	—	—
5L/P63 (P387)	S	S	S	N(60)	N(56)	N(51)	—	—	—	—	—
5L/P64 (P389)	S	S	S	S	S	N(56)	—	—	—	—	—
5L/P65 (P391)	S	S	S	S	S	N(56)	—	—	—	—	—
5L/P66 (P398)	S	S	S	N(66)	N(57)	N(59)	—	—	—	—	—

master	ε^{-5}	ε^{-4}	ε^{-3}	ε^{-2}	ε^{-1}	ε^{0}	ε^{1}	ε^{2}	ε^{3}	ε^{4}	ε^{5}
5L/P67 (P399)	S	S	S	S	S	N(57)	—	—	—	—	—
5L/P68 (P400)	S	S	S	S	S	N(33)	—	—	—	—	—
5L/P69 (P413)	S	S	S	S	N(83)	N(74)	—	—	—	—	—
5L/P70 (P418)	S	S	S	S	N(82)	N(74)	—	—	—	—	—
5L/P71	S	S	S	S	N(57)	N(58)	—	—	—	—	—
5L/P72	S	S	S	N(64)	N(57)	N(57)	—	—	—	—	—
5L/P73 (P422)	S	S	S	N(82)	N(82)	N(74)	—	—	—	—	—
5L/P74 (P424)	S	S	S	S	N(84)	N(75)	—	—	—	—	—
5L/P75 (P426)	S	S	S	S	N(58)	N(57)	—	—	—	—	—
5L/P76 (P427)	S	S	S	S	N(83)	N(75)	—	—	—	—	—
5L/P77 (P428)	S	S	S	S	S	N(57)	—	—	—	—	—
5L/P78 (P429)	S	S	S	S	S	N(74)	—	—	—	—	—
5L/P79 (P432)	S	S	S	S	S	N(62)	—	—	—	—	—
5L/P80 (P433)	S	S	S	S	S*(2395)[p]	N(47)	—	—	—	—	—
5L/P81	S	S	S	N(65)	N(57)	N(57)	—	—	—	—	—
5L/P82 (P435)	S	S	S	N(48)	N(47)	N(47)	—	—	—	—	—
5L/P83 (P436)	S	S	S	S	N(82)	N(75)	—	—	—	—	—
5L/P84 (P440)	S	S	S	N(47)	N(47)	N(47)	—	—	—	—	—
5L/P85 (P441)	S	S	S	S	S	N(39)	—	—	—	—	—
5L/P86 (P442)	S	S	S	S	S*(2395)[p]	S*(2395)[p]	—	—	—	—	—
5L/P87 (P443)	S	S	S	S	S	N(57)	—	—	—	—	—
5L/P88 (P444)	S	S	S	S	S	N(44)	—	—	—	—	—
5L/P89 (P445)	S	S	S	S	S	N(32)	—	—	—	—	—
5L/P90 (P446)	S	S	S	S	S	N(35)	—	—	—	—	—
5L/P91 (P447)	S	S	S	S	S	S	—	—	—	—	—
5L/P92 (P448)	S	S	S	S	N(57)	N(57)	—	—	—	—	—
5L/P93 (P449)	S	S	S	S	S	S	—	—	—	—	—
5L/P94 (P451)	S	S	S	S	N(57)	N(57)	—	—	—	—	—
5L/P95 (P453)	S	S	S	S	N(83)	N(75)	—	—	—	—	—
5L/P96 (P454)	S	S	S	S	S*(2600)[p]	S*(2348)[p]	—	—	—	—	—
5L/P97 (P455)	S	S	S	N(47)	N(47)	N(47)	—	—	—	—	—
5L/P98 (P456)	S	S	S	S	S*(2600)[p]	N(47)	—	—	—	—	—
5L/P99 (P457)	S	S	S	S	S	N(53)	—	—	—	—	—
5L/P100 (P460)	S	S	S	S	N(82)	N(75)	—	—	—	—	—
5L/P101 (P462)	S	S	N(48)	N(48)	N(48)	N(48)	—	—	—	—	—
5L/P102 (P463)	S	S	S	N(47)	N(47)	N(47)	—	—	—	—	—
5L/P103 (P464)	S	S	S	S	S	N(48)	—	—	—	—	—
5L/P104 (P465)	S	S	S	S	S*(2600)[p]	N(47)	—	—	—	—	—
5L/P105 (P466)	S	S	S	S	S	N(55)	—	—	—	—	—
5L/P106 (P470)	S	S	S	S	N(82)	N(74)	—	—	—	—	—
5L/P107 (P472)	S	S	S	S	N(47)	N(46)	—	—	—	—	—
5L/P108 (P477)	S	S	S	S	N(57)	N(57)	—	—	—	—	—
5L/P109 (P478)	S	S	S	S	S*(5363)	N(75)	—	—	—	—	—
5L/P110 (P480)	S	S	S	S	N(82)	N(75)	—	—	—	—	—
5L/P111 (P483)	S	S	S	S	S	N(56)	—	—	—	—	—
5L/P112 (P484)	S	S	S	S	S	N(55)	—	—	—	—	—
5L/P113	S	S	S	S	N(56)	N(56)	—	—	—	—	—
5L/P114 (P501)	S	S	S	S	S*(5363)	N(56)	—	—	—	—	—
5L/P115 (P505)	S	S	S	S	S	N(43)	—	—	—	—	—
5L/P116 (P506)	S	S	S	S	S*(5363)	N(57)	—	—	—	—	—
5L/P117 (P509)	S	S	S	S	N(35)	N(35)	—	—	—	—	—
5L/P118 (P511)	S	S	S	S	N(56)	N(57)	—	—	—	—	—
5L/P119 (P519)	S	S	S	S	S*(5527)	S*(5525)	—	—	—	—	—
5L/P120 (P530)	S	S	S	S	S*(5363)	S*(5361)	—	—	—	—	—
5L/P121 (P533)	S	S	S	S	N(82)	N(74)	—	—	—	—	—
5L/P122 (P536)	S	S	S	S	N(81)	N(74)	—	—	—	—	—
5L/P123 (P539)	S	S	S	S	S	N(62)	—	—	—	—	—
5L/P124 (P550)	S	S	S	S	S*(5363)	N(75)	—	—	—	—	—
5L/P125 (P551)	S	S	S	S	S	N(40)	—	—	—	—	—
5L/P126 (P553)	S	S	S	S	S	N(61)	—	—	—	—	—
5L/P127 (P560)	S	S	S	S	N(82)	N(74)	—	—	—	—	—
5L/P128	S	S	S	S	N(56)	N(56)	—	—	—	—	—
5L/P129 (P631)	S	S	S	S	S	N(63)	—	—	—	—	—
5L/P130 (P632)	S	S	S	S	S	N(40)	—	—	—	—	—
5L/P131 (P634)	S	S	S	S	S	N(54)	—	—	—	—	—

[p] Two-profile constants (digits; independent confirmation): vakint::D5bar3 (9924 d; independent implementations agree to 5060 d): 4 orders; vakint::PR0D2Eps1 (2395 d; independent confirmation 48.5 d): 5 orders; vakint::PR0D2Eps2 (2396 d; independent confirmation 51.2 d): 2 orders; vakint::PR12D2Eps2 (2600 d; independent confirmation 48.7 d): 6 orders; vakint::PR12D2Eps3 (2348 d; independent confirmation 421 d): 2 orders; vakint::PR15D2Eps1 (2805 d; independent confirmation 794.27 d): 6 orders; vakint::PR15D2Eps2 (2601 d; independent confirmation 794.42 d): 9 orders; vakint::PR15D2Eps3 (2351 d; independent confirmation 793.39 d): 2 orders; vakint::PR5D2Eps4 (9923 d; independent implementations agree to 5060 d): 4 orders. [n] Numerical mode uses the best payload of a master's coefficient group (same exact basis expression), here a validated numerical value: 4L/P17 ε^0 S*(5363) → 20000 d; 4L/P19 ε^0 S*(5363) → 20000 d; 4L/P52 ε^0 S*(5363) → 20000 d; 4L/P73 ε^0 S*(5363) → 20000 d; 4L/P74 ε^0 S*(5363) → 20000 d.

Precision table 2. The L-loop result — class and digits reachable, worst case over the admitted inputs, per Laurent order of the result, with the spurious-pole orders the row notes state (measured at 1L-4L; at 5L proven on the held integrals and sampled outside them, but not in the top sector: a measured top-sector input is folded in, note k; letters: notes below).

L	ε^{-5}	ε^{-4}	ε^{-3}	ε^{-2}	ε^{-1}	ε^{0}	ε^{1}	ε^{2}	ε^{3}	ε^{4}	ε^{5}
1L[a]					S	S	S	S	S	S	S
2L[a]				S	S	S	S	S	S	—[b] some inputs: S	—
3L[a]			S	S	S	S	S	—[c] some inputs: S	—[d] some inputs: S	—[e] some inputs: S	—
4L[a]		S	S	S	S* 20000 digits[f]	S* 2395 digits[g]	—[h] some inputs: S*(2348)	—[i] some inputs: S	—[j] some inputs: S	—	—
5L[k,l]	S	N 65 digits[m] others: S	N 47 digits[n] others: N(48)	—[o] some inputs: N(47)	—[p] some inputs: N(35)	—[q] some inputs: N(32)	—	—	—	—	—

a 1L-4L: spurious-pole orders p measured on the shipped caches and samples (direct route up to 1L 0, 2L 0, 3L 1, 4L 2), not proven.
b 2L ε^{4}: inputs reaching 2L/P1 get MissingMasterOrders; others S.
c 3L ε^{2}: inputs reaching 3L/P1, 3L/P3, 3L/P6 +9 get MissingMasterOrders; others S.
d 3L ε^{3}: inputs reaching 3L/P1, 3L/P2, 3L/P3 +14 get MissingMasterOrders; others S.
e 3L ε^{4}: inputs reaching 3L/P1, 3L/P2, 3L/P3 +35 get MissingMasterOrders; only inputs with lower spurious poles, e.g. 3L/P13, 3L/P21, 3L/P28: S.
f 4L ε^{-1} S*(20000): K5Moment(1), K5Moment(3) via $1/\varepsilon^{1}$ poles.
g 4L ε^{0} S*(2395): PR0D2Eps1 (two profiles of one code).
h 4L ε^{1}: inputs reaching 4L/P6, 4L/P7, 4L/P9 +18 get MissingMasterOrders; so do 4L/P5, 4L/P8, 4L/P15 +5 combined with 4L/P35, 4L/P51; others S*(2348) (4L/P70 ε^{1}.
i 4L ε^{2}: inputs reaching 4L/P1, 4L/P3, 4L/P5 +59 get MissingMasterOrders; others S.
j 4L ε^{3}: inputs reaching 4L/P1, 4L/P2, 4L/P3 +71 get MissingMasterOrders; only inputs with lower spurious poles, e.g. 4L/P2, 4L/P21, 4L/P37 +1: S.
k 5L: p = 0 (pole-free basis BASIS-v2) proven on the 2336 held integrals and sampled outside them. Not so in the top sector: the measured input P105 + dot has p = 1 on 59 5L masters and up to p = 3 on its 4L product factors; other top-sector inputs are not validated.
l 5L results hold products of 1L-4L masters: a factor of L_f loops is needed through ε^{k+5-L_f+p} (p measured on the factor images).
m 5L ε^{-4} N(65): P105 + dot needs 5L/P6 (P60) ε^{-3}; others S.
n 5L ε^{-3} N(47): P105 + dot needs 5L/P50 (P357) ε^{-2}, 5L/P84 (P440) ε^{-2}; others N(48).
o 5L ε^{-2}: P105 + dot gets MissingMasterOrders; others N(47) (5L/P102 (P463) ε^{-2}, 5L/P50 (P357) ε^{-2}.
p 5L ε^{-1}: P105 + dot gets MissingMasterOrders; others N(35) (5L/P117 (P509) ε^{-1}.
q 5L ε^{0}: P105 + dot gets MissingMasterOrders; others N(32) (5L/P89 (P445) ε^{0}.

Method. A coefficient is limited by its fewest-digit finite source (4L constants, numerical master values); the result at ε^{k} needs every master m through ε^{k+p_m} (p_m: spurious-pole order of its reduction coefficient), on vakint's basis-alias route first, else the direct route; a needed order the data do not hold gives MissingMasterOrders (no value, never a wrong one). RustRed backend only. Full notes, method, finite-digit periods, the numerical identities behind some S and the PSLQ identifications: `crates/vakint/docs/benchmark/data/PRECISION-TABLES.md`.